TP1 - Exercise 1: Answers

Q2. What each output tells us

info() shows the number of rows, the column names, the data type of each column and the number of non-null values (so we can see if values are missing).

describe() gives, for each numeric column, the count, mean, standard deviation, minimum, quartiles (25%, 50%, 75%) and maximum.

shape gives the size of the dataset as (number of rows, number of columns).

columns gives the names of the columns.

head() shows the first 5 rows, to have a quick look at what the data looks like.

Q5. Why make a copy?

A copy lets us test preprocessing (scaling, encoding, deleting rows...) without changing the original data. If something goes wrong, we still have the original DataFrame and we do not need to reload the file.

Q7. Features and target

The input features are sepal_length, sepal_width, petal_length and petal_width. The target is the class label, species. We use stratify=y so that the three classes keep the same proportions in the training set and in the test set.

Discussion: why keep the test set separate?

The test set plays the role of new data that the model has never seen. It must stay separate until the final evaluation to get an honest estimate of the performance.

If information from the test set influences the preprocessing (for example, computing the mean for scaling on all the data) or the model selection, then the test data indirectly influences the model. This is called **data leakage**. The score becomes too optimistic, and the model will perform worse on real new data.